22MA201 & ENGINEERING ~~CHEMISTRY~~ MATHEMATICS - II

1. Sol

$\vec{r}(t) = t\hat{i} + t^2\hat{j}$

velocity vector, $\vec{v}(t) = \frac{d\vec{r}}{dt}$

$\vec{v}(t) = \hat{i} + 2t\hat{j}$

Interpretation:

* constant motion in x direction, Increasing motoin in y-direction
* The path is parabolic

2.

Scalar line Integral	Vector line Integral
* Integrates a scalar function over a curve $\int_C f\,ds$	* Integrates a vector field along a curve $\int_C \vec{F}\cdot d\vec{r}$
* magnitude only	* Magnitude and direction

3. Sol

$\vec{F} = (x+y)\hat{i} + (x-y)\hat{j}$ — (1)

Green's Theorem:

$$\oint_C M(x,y)\,dx + N(x,y)\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$$

$\therefore M = x+y$, $N = x-y$ [From eqn (1)

$\frac{\partial N}{\partial x} = 1$ $\frac{\partial M}{\partial y} = 1$

$$\oint_C M(x,y)\,dx + N(x,y)\,dy = \iint_R \left(\frac{dN}{dx} - \frac{\partial M}{\partial y}\right) dA$$

$$= \iint_R (1-1)\,dA = \iint_R 0\,dA$$

$$= 0$$

4. Sol
A positive value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} > 0$ indicates positive circulation (counterclockwise rotation) of the field

5. The line integral of a force feild along a curve gives the work done by the force on the particle

$$\int_C \vec{F} \cdot d\vec{r}$$

Interpretation:
It depends on both the force and the path followed

6. Path ~~independen~~ independence means the work done depends only on the start and end points, not on the path followed.

7. Sol

$$\vec{F} = y\hat{i} + x\hat{j}$$

$$P = y, \quad Q = x$$

using Green's theorem integrant:

$$\oint_C M(x,y)\,dx + N(x,y)\,dy = \iint_R \left(\frac{dQ}{dx} - \frac{dP}{dy} \right) dA$$

$$= \iint_R (1-1)\,dA = \iint_R 0\,dA$$

$$= 0$$

Hence, It is indicating there is no circulation or rotation in the fluid flow.

8. Sol

$$\vec{F} = (x+y)\hat{i} + (2x-y)\hat{j}$$

$P = x+y$
$Q = 2x-y$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial(2x-y)}{\partial x} - \frac{\partial(x+y)}{\partial y}$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2-1 = 1$$

$\therefore$ The required value is 1

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9. Sol

To evaluate : $\iint_S \vec{F}\cdot\hat{n}\,ds$

Given : $\vec{F} = x^2\hat{i} + yz\hat{j} + z^2\hat{k}$ $\quad \hat{n} = \hat{j}$ $\quad y=1$, $0 \le x \le 2$, $0 \le z \le 1$

$$\vec{F}\cdot\hat{n} = (x^2\hat{i} + yz\hat{j} + z^2\hat{k})\cdot\hat{j} = yz \Rightarrow \text{For } y=1$$

$$\vec{F}\cdot\hat{n} = z$$

$ds = dx\,dz$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \int_0^2\int_0^1 z\,dz\,dx = \int_0^2 \left[\frac{z^2}{2}\right]_0^1 dx = \int_0^2 \frac{1}{2}\,dx$$

$$= \left[\frac{x}{2}\right]_0^2 = \left[\frac{2}{2} - \frac{0}{2}\right]$$

$$= 1$$

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10. Sol

Stroke's Theorem:

$$\int_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot\hat{n}\,ds \quad \text{—(1)}$$

Given:

$\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$ $\quad x=0$, $0 \le y \le 1$, $0 \le z \le 1$

For yz plane, unit normal vector, $\hat{n} = \hat{i}$

$ds = dy\,dz$

$$\nabla\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \left(\frac{\partial y}{\partial y} - \frac{\partial x}{\partial z}\right)\hat{i} + \left(\frac{\partial z}{\partial z} - \frac{\partial y}{\partial x}\right)\hat{j} + \left(\frac{\partial x}{\partial x} - \frac{\partial z}{\partial y}\right)\hat{k}$$

$$= \hat{i} + \hat{j} + \hat{k}$$

$(\nabla\times\vec{F})\cdot\hat{n} = (\hat{i}+\hat{j}+\hat{k})\cdot(\hat{i}+0\hat{j}+0\hat{k})$

$(\nabla\times\vec{F})\cdot\hat{n} = 1$

From eqn (i)

$$\int_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot\hat{n}\,ds$$

$$= \int_0^1\int_0^1 1\,dy\,dz = \int_0^1 [y]_0^1\,dz = \int_0^1 1\,dz = [z]_0^1$$

$$\int_C \vec{F}\cdot d\vec{r} = 1$$

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11. Sol

To Evaluate: $\iint_S \vec{F}\cdot\hat{n}\,ds$

$x=2$

$0\le y\le 1$, $0\le z\le 2$

$\therefore \hat{n}=\hat{i}$

$\vec{F} = 2xy\hat{i} + yz\hat{j} + xz\hat{k}$

$\vec{F}\cdot\hat{n} = (2xy\hat{i}+yz\hat{j}+xz\hat{k})\cdot(\hat{i}+0\hat{k}+0\hat{j})$

$\vec{F}\cdot\hat{n} = 2xy$

$x=2$

$\therefore \vec{F}\cdot\hat{n} = 4y$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \int_0^2\int_0^1 4y\,dy\,dz = \int_0^2 4\left[\frac{y^2}{2}\right]_0^1 dz = \int_0^2 2\,dz = 2[z]_0^2$$

$= 2(2-0)$

$= 4$ //

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12. Sol

Stroke's Theorem:

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot\hat{n}\,ds$$

Given:

$\vec{F} = xy\hat{i} + yz\hat{j} + 2x\hat{k}$

$y = 0$

$0 \le x \le 1$

$0 \le z \le 1$

$\therefore \hat{n} = \hat{j}$

$ds = dx\,dz$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & yz & 2x \end{vmatrix}$$

$$= \left(\frac{\partial(2x)}{\partial y} - \frac{\partial(yz)}{\partial z}\right)\hat{i} + \left(\frac{\partial(xy)}{\partial z} - \frac{\partial(2x)}{\partial x}\right)\hat{j} + \left(\frac{\partial(yz)}{\partial x} - \frac{\partial(xy)}{\partial y}\right)\hat{k}$$

$$= (0-y)\hat{i} + (0-z)\hat{j} + (0-x)\hat{k}$$

$$\nabla \times \vec{F} = -y\hat{i} - z\hat{j} - x\hat{k}$$

$$(\nabla \times \vec{F})\cdot\hat{n} = (-y\hat{i} - z\hat{j} - x\hat{k})\cdot(0\hat{i} + \hat{j} + 0\hat{k})$$

$$= -z$$

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S (-z)\,ds = \int_0^1\int_0^1 (-z)\,dx\,dz$$

$$= \int_0^1 -z[x]_0^1\,dz = \int_0^1 -z\,dz = -\left[\frac{z^2}{2}\right]_0^1 = -\frac{1}{2}$$

$$\oint_C \vec{F}\cdot d\vec{r} = \frac{-1}{2}$$

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13. Sol

Green's theorem:

$$\oint_C M(x,y)dx + N(x,y)dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)dA$$

Given

$\vec{F} = y\hat{i} + x\hat{j}$

$0 \le x \le 1$, $0 \le y \le 1$

Here

$N = x$

$M = y$

$$\frac{\partial N}{\partial x} = 1 \qquad \frac{\partial M}{\partial y} = 1$$

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

$$= \iint_R (1-1)\, dA = \iint 0\, dA$$

$$= 0$$

The work done is zero

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14. Sol

Divergence Theorem:

$$\iint_S \vec{F}\cdot\hat{n}\, ds = \iiint (\nabla\cdot\vec{F})\, dv$$

$$\vec{F} = (x^2-yz)\hat{i} + (y^2-xz)\hat{j} + (z^2-xy)\hat{k}$$

$0 \le z \le c$ $\qquad dv = dz\,dy\,dx$

$0 \le x \le a,$

$0 \le y \le b,$

$$\nabla\cdot\vec{F} = \frac{\partial}{\partial x}(x^2-yz) + \frac{\partial}{\partial y}(y^2-xz) + \frac{\partial}{\partial z}(z^2-xy)$$

$$= 2x + 2y + 2z$$

$$\iint_S \vec{F}\cdot\hat{n}\, ds = \iiint_V (2x+2y+2z)\, dv$$

$$= 2\iiint_V (x+y+z)\, dv$$

$$= 2\int_0^a\int_0^b\int_0^c (x+y+z)\, dz\,dy\,dx$$

$$= 2\left[\frac{a^2bc}{2} + \frac{ab^2c}{2} + \frac{abc^2}{2}\right]$$

$$= a^2bc + ab^2c + abc^2$$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = abc\,(a+b+c)$$

Interpretation:

since the flux value is positive

$$abc\,(a+b+c) > 0$$

there is a net outward flow of heat from the rectangular region

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15. Sol

Given:

$$\vec{F} = 2x\hat{i} + y\hat{j} + z\hat{k}$$

$$x = 1,\ 0 \le y \le 2,\ 0 \le z \le 1 \qquad \therefore \hat{n} = \hat{i}$$

To find: $\iint_S \vec{F}\cdot\hat{n}\,ds$.

$$\vec{F}\cdot\hat{n} = (2x\hat{i} + y\hat{j} + z\hat{k})\cdot(\hat{i})$$

$$= 2x$$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \int_0^1\int_0^2 2\,dy\,dz = \int_0^1 2[y]_0^2\,dz = 2\int_0^1 [2-0]\,dz$$

$$= 4\int_0^1 dz = 4[z]_0^1 = 4(1-0)$$

$$= 4$$

$\therefore$ The total flux through the surface is 4 units

16. sol

$\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ $dv = dx\,dy\,dz$

$0 \leq x, y, z \leq 1$

Divergence Theorem: $\iint_S \vec{F}\cdot\hat{n}\,ds = \iiint_V (\nabla\cdot\vec{F})\,dV$

$$\nabla\cdot\vec{F} = \frac{\partial(4xz)}{\partial x} + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(yz)$$

$$= 4z - 2y + y$$

$$\nabla\cdot\vec{F} = 4z - y$$

$$\therefore \iint_S \vec{F}\cdot\hat{n}\,ds = \iiint_V (4z-y)\,dv = \iiint_V (4z-y)\,dx\,dy\,dz$$

$$= \int_0^1\int_0^1\int_0^1 (4z-y)\,dx\,dy\,dz$$

$$= \int_0^1\int_0^1 (4z-y)\,dy\,dz = \int_0^1 (4z-y)\,dz$$

$$= \left[2z^2 - \frac{z}{2}\right]_0^1 = 2 - \frac{1}{2}$$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \frac{3}{2}$$

Interpretation:

Since the flux is positive

$\frac{3}{2} > 0$

There is a net outward heat flow from the cubic region

17. Sol

$f(z) = z^2$

$f(z) = (x+iy)^2$

$f(z) = x^2 - y^2 + 2ixy$

CR equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x} = 2, \quad \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = 2y$$

$\therefore$ It obeys CR equations

Therefore, $f(z) = z^2$ is differentiable at $z = 1+i$

18. Sol

$w = z^2$

* For each complex number z, there is only one unique value of w. Hence, the function does not produce multiple outputs for the same input

* $\therefore$ $w = z^2$ is a single valued function

19. Sol Given: $v(x,y) = 2xy$

To find: $u(x,y)$

Milne-Thompson method:

$$f'(z) = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$$

$$\frac{\partial v}{\partial y} = 2x, \quad \frac{\partial v}{\partial x} = 2y$$

$$f'(z) = 2x + i2y = 2(x+iy)$$

$$f'(z) = 2z \qquad [\because z = x+iy$$

$\therefore f(z) = \int 2z\, dz$

$f(z) = z^2 + C$

$z^2 = (x+iy)^2$

$z^2 = \underbrace{x^2 - y^2}_{u} + i\underbrace{(2xy)}_{v}$

$\therefore \boxed{u(x,y) = x^2 - y^2}$

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20. Sol

Given : $u(x,y) = x^2 + 2xy$

To find : $f'(z)$

Using Milne - Thomson Method,

~~$f'(z) = u_x(z,0) - i(z$~~

$f'(z) = u_x(z,0) - i u_y(z,0)$

$u_x = 2x + 2y$

$u_y = 2x$

put $y = 0$, $x = z$

$\therefore u_x = 2z$

$u_y = 2z$

$\therefore f'(z) = 2z - i(2z)$ $\boxed{f'(z) = 2z(1-i)}$

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21. Sol

* If complex function $f(z)$ is differentiable, then it is continuous
* But a continuous function need not be differentiable
* Hence, Differentiability $\Rightarrow$ continuity
* But the converse is not always true

22. Sol

$w = \sqrt{z}$

*For one ~~vol~~ value of z, the function gives two values of w (positive and negative roots)

* So, $w = \sqrt{z}$ is multi-value function

23. Sol

Given: $u(x,y) = e^x \cos y$

To find: $f(z)$ ~~[illegible]~~

Milne-Thomson method:

$$f'(z) = u_x - iu_y$$

$$u_x = e^x \cos y$$

$$u_y = -e^x \sin y$$

$$f'(z) = e^x \cos y + ie^x \sin y$$

$$= e^x(\cos y + i\sin y)$$

$$f'(z) = e^{x+iy} = e^z \Rightarrow f'(z) = e^z$$

$$f(z) = \int e^z dz$$

$$\boxed{f(z) = e^z + C}$$

24. Sol

Given: $v(x,y) = x^2 - y^2$

to find: $f'(z)$

Milne-Thomson method:

$$f'(z) = v_y(z,0) + iv_x(z,0)$$

$$v_x = 2x$$

$$v_y = -2y \qquad f'(z) = 0 + i(2z)$$

Put $y=0$, $x = z$

$\therefore v_x = 2z \qquad \boxed{f'(z) = 2iz}$

$v_y = 0$